

RV COLLEGE OF ENGINEERING® , BENGALURU-59
(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



**Traffic Monitoring System Using YOLO and
Deep SORT**

**ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
(IS353IA)**

V SEMESTER

OPEN-ENDED PROJECT REPORT

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in
Computer Science and Engineering**

2024-2025

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CERTIFICATE

Certified that the **Artificial Intelligence and Machine Learning Open-Ended Project Work** titled “**Traffic Monitoring System Using YOLO and Deep SORT**” is carried out by **Chandan Kumar K R (1RV23CS402, Gopinath Ramaje (1RV23CS404) and Kiran H R (1RV23CS405)** who are Bonafide student/s of RV College of Engineering, Bengaluru, in partial fulfillment for the **Internal Assessment of Course: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (IS353IA)** during the year 2024-2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report.

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DECLARATION

We, **Chandan Kumar K R (1RV23CS402), Gopinath Ramaje (1RV23CS404) and Kiran H R (1RV23CS405)** the students of Fifth Semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru hereby declare that project titled **“Traffic Monitoring System Using YOLO and Deep SORT”** has been carried out by us and submitted in partial fulfillment for the **Internal Assessment of the Course: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (IS353IA)** during the academic year 2024-2025. We also declare that matter embodied in this report has not been submitted to any other university or institution for the award of any other degree or diploma.

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ABSTRACT

With increasing urbanization and rising vehicle numbers, traffic congestion and road safety have become critical concerns. This project presents a real-time traffic monitoring system that utilizes YOLOv8 for accurate object detection and Deep SORT for efficient vehicle tracking. The system detects and tracks vehicles in real-time, classifies them based on type, estimates their speeds, and identifies potential speed limit violations. By leveraging deep learning and computer vision, the system overcomes challenges such as occlusions, varying lighting conditions, and high-speed vehicle movements, ensuring reliable monitoring in complex urban environments.

The system offers key functionalities, including real-time vehicle classification, speed estimation, congestion analysis, and traffic data logging for long-term insights. A user-friendly interface provides visual analytics, displaying traffic patterns such as vehicle counts, speed distributions, and density trends. These insights support urban planners, traffic management authorities, and law enforcement agencies in making data-driven decisions to optimize road safety and efficiency. The system significantly reduces manual intervention, enhances accuracy, and enables proactive traffic regulation.

Future enhancements include automated violation detection (e.g., red-light running, lane violations, pedestrian movement tracking) and IoT-based adaptive traffic control, which dynamically adjusts traffic signals based on real-time congestion levels. By integrating AI-driven monitoring and analytics, this solution contributes to smarter, safer, and more efficient transportation networks, aiding cities in achieving sustainable and intelligent urban mobility.

ACRONYMS

- YOLO – You ONLY Look ONCE
- SORT– SIMPLE ONLINE AND REALTIME TRACKER
- CNN – CONVOLUTIONAL NEURAL NETWORK
- CV – COMPUTER VISION
- DEEP SORT – DEEP SIMPLE ONLINE AND REALTIME TRACKER
- FPS – Frames Per Second RGB
- RTSP – REAL – TIME STREAMING PROTOCOL

TABLE OF CONTENTS

Page No

Abstract

List of Tables

List of Figures

Chapter 1

Introduction 1

1.1. State of Art Developments 2

1.2. Motivation 4

1.3. Problem Statement 4

1.4. Objectives 4

1.5. Scope 5

1.6. Methodology 5

Chapter 2

Overview of AI and ML Component in the Problem Domain 7

2.1. Introduction

2.2. Relevant Technical and Mathematical Details 8

2.3 Summary 10

Chapter 3

Software Requirements Specification of Traffic Monitoring System using Yolo and Deep Sort 12

3.1 Software Requirements 13

Chapter 4

Design of Traffic Monitoring System using Yolo and Deep Sort 14

4.1 System Architecture 15

4.2 Functional Description of the Modules 17

4.2.1. Input Module 17

4.2.2. Object Detection Module 17

4.2.3. Multi-Object Tracking Module 18

Chapter 5	
Implementation of Traffic Monitoring System using Yolo and Deep Sort	20
5.1. Programming Language Selection	21
5.2. Platform Selection	21
Chapter 6	
Experimental Results and Analysis of Traffic Monitoring System using Yolo and Deep Sort	23
6.1. Evaluation Metrics	24
6.2. Experimental Dataset	24
6.3. Performance Analysis	25
Chapter 7	
Conclusion and Future Enhancement	27
7.1. Limitations of the Project	28
7.2. Future Enhancements	28
References	30
Appendices	
Appendix 1: Screenshots	34

List of Figures

FIGURE NO.	TITLE	PAGE NO.
2.1	Bird's Eye View	10
4.1	Traffic Monitoring System Workflow	16
6.1	Real Time Traffic Monitoring System	26